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Excel Syllabus

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Meaning

Microsoft excel is a spreadsheet software used to store, organize, calculate, analyze and visualize data. It is a beginner friendly

workbook : a workbook is the entire excel file with (.xlsx) extension

worksheet : a worksheet is an individual sheet inside the workbook.

row : run horizontally and are numbered:
1, 2, 3, 4...

columns: runs vertically and are named A, B, C, D
cell: The intersection of row and column is called a cell

Why we need excel

- * stores large amounts of data
- * clean data, perform calculation
- * perf analyze data, create charts
- * finds patterns and trends.
- * no need of coding

Types of data in Excel

1. Text data : contains letters, word, characters
2. Number data : no. are used for calculation.
3. Date data : date represents a specific day, month, year
4. Time data : Represents a specific time
5. Currency data : represents money value
6. Boolean data : logical data with two possible values. (T/F, 0/1)

Important excel features.

1. Data cleaning
 - * This finds and corrects problems in raw data
 - * removing duplicate records.

- * corrects missing values, spellings, extra spaces.
- * Incorrect data formats.

2. formulas and functions.

A formula is an expression we create to perform a calculation.

SUM, AVERAGE, COUNT, COUNTA, MAX, MIN

3. Data visualization.

Representing data visually using charts and graphs like bar graph, pie chart

4. Pivot table.

Allows to summarize a large amt of data quickly. very important

5. Sorting and filtering

- * arranging data smallest to largest, largest to smallest
- * filtering means showing only the data we want while hiding the rest temporarily

6. Data validation

This allows and controls what a users are

allowed to enter in a cell.

Advantage.

- Easy to learn
- powerful calculation
- Data visualization & Data analysis
- widely used & flexible

Disadvantage.

- can become difficult with huge datasets
- Manual errors
- collaboration limitations
- Security concerns
- Repetitive work.

21/9/26 Functions

Set of instructions / set of code is already predefined is called function.

3 main concept of function:

- 1) function name
- 2) no. of inputs
- 3) return value

Excel functions.

1. **upper()** : This function converts all letters in a text into capital letters

syntax : =UPPER(text)

Ex : =UPPER(A2) , UPPER("Sanjana")

O/P : SANJANA

2. **lower()** : Converts all letters in a text into small letters

syntax : =LOWER(text)

Ex : =LOWER(A2) , =LOWER("SANJANA")

O/P : sanjana.

3. **Proper()** : This converts the first letter of each word into capital letter and remaining letters into small letters.

syntax : =PROPER(text)

Ex : =PROPER(A2) , =PROPER("SANJANA")

O/P : Sanjana

4. **LEN()**: This counts the no. of characters in a text and also spaces are counted as a char

Syntax: =LEN(text)

Ex: =LEN(A2), =LEN("sanjana")

OP: 7

5. **MID()**: This function extracts a specific no of characters from the middle of a text, starting from a position we specify

Syntax: =MID(text, start-num, num-chars)

Ex: =MID(A2, 2, 3), =MID("sanjana", 2, 3)

OP: anj

6. **LEFT()**: This extracts a specified no of characters from the left side of the text

Syntax: =LEFT(text, num-char)

Ex: =LEFT(A2, 3), =LEFT("sanjana", 3)

OP: san

7. **RIGHT()**: extracts a specified no. of characters from the right side of a text

Syntax : =RIGHT (text , num - chars)

Ex : =RIGHT (A2 , 3) , =RIGHT ("sanjana" , 3)

OP : ana

8. Dividing the text into 1st half and 2nd half
→ used when we need to split a text into 2 parts.

for even numbers : ANUSHA (6 chars)

1st half syntax : =LEFT (text , LEN (text) / 2)

Ex : =LEFT (A2 , LEN (A2) / 2 , 0)

OP : ANU

2nd half syntax : =RIGHT (text , LEN (text) / 2)

Ex : =RIGHT (A2 , LEN (A2) / 2 , 0)

OP : SHA

for odd numbers : SANJANA (7 chars)

when the text has an odd no. of chars, it cannot be divided into equal halves

so, ROUNDUP() function should be used in any one of the half

1st half syntax : =LEFT (text , ROUNDUP (LEN (text) / 2 , 0))

Ex : =LEFT (A2 , ROUNDUP (LEN (A2) / 2 , 0))

OP : SANJ

2nd half syntax : $=\text{RIGHT}(\text{text}, \text{LEN}(\text{text})/2)$

EX : $=\text{RIGHT}(A2, \text{LEN}(A2)/2)$

OP : ANA